

CAK2HAB3 - Dasar Kecerdasan Artifisial

Searching: Heuristic (Informed) Search

Lecturer Team



Outline

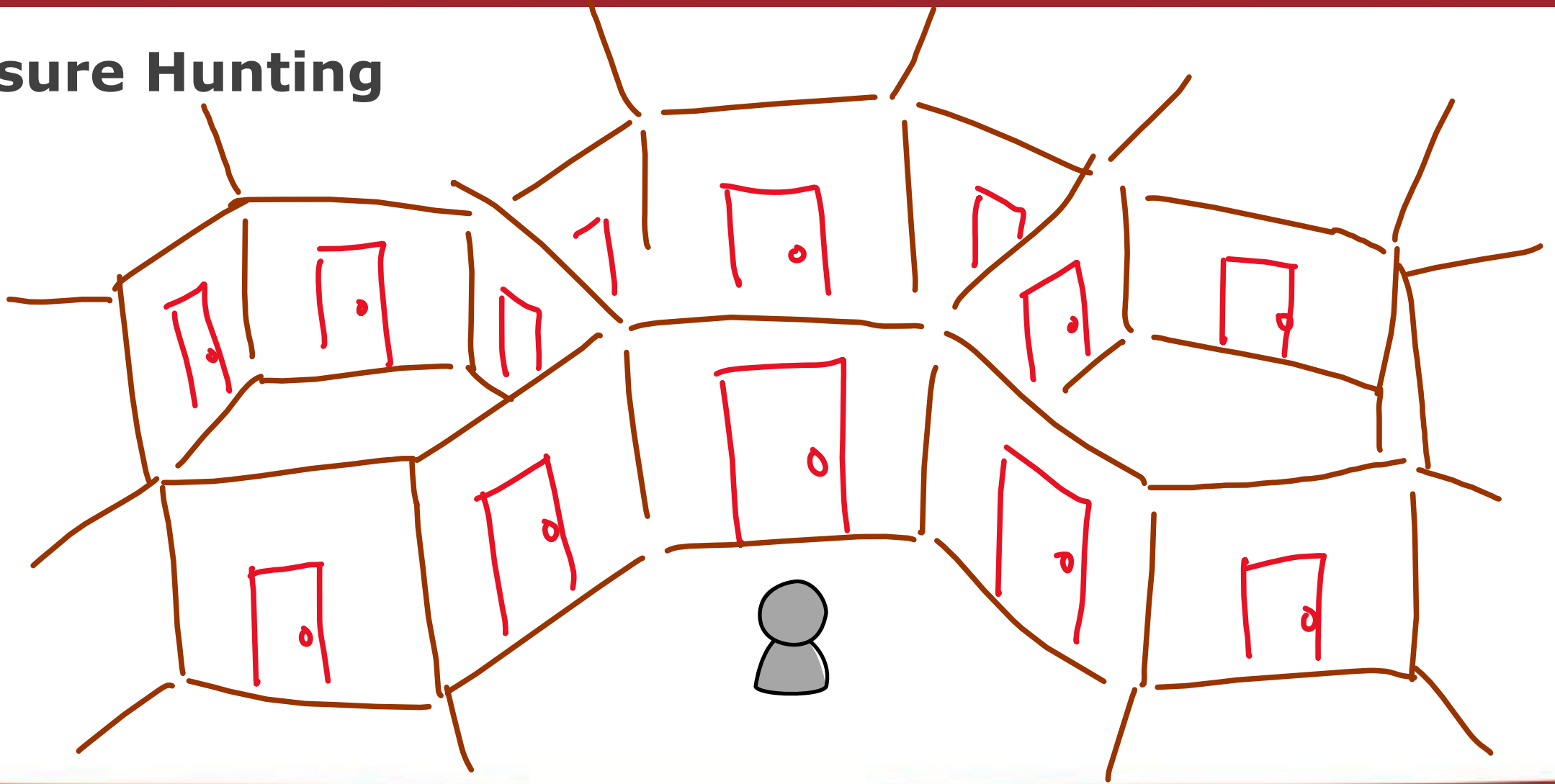
- ▶ Heuristic Function



Recall: Blind Search

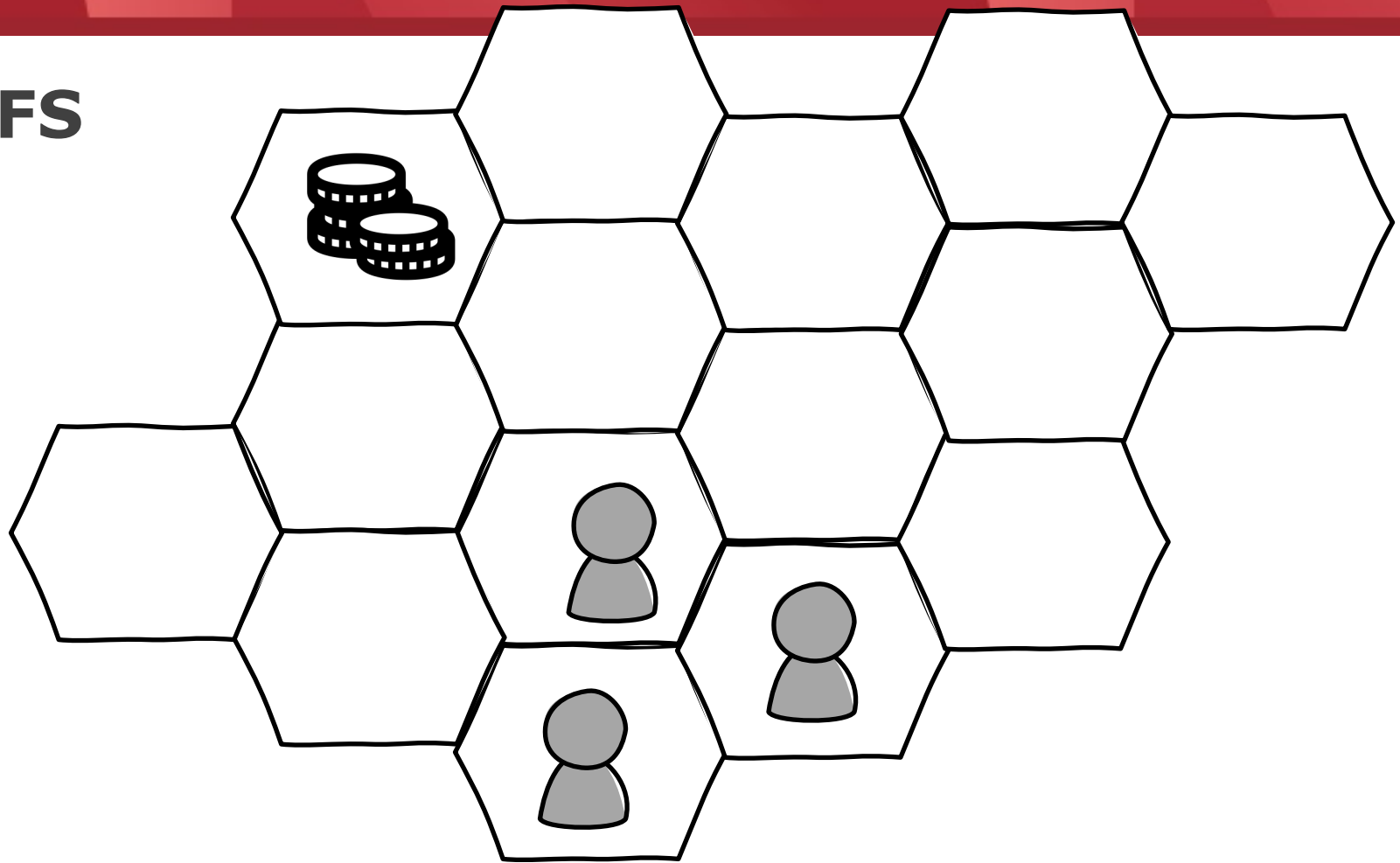


Treasure Hunting



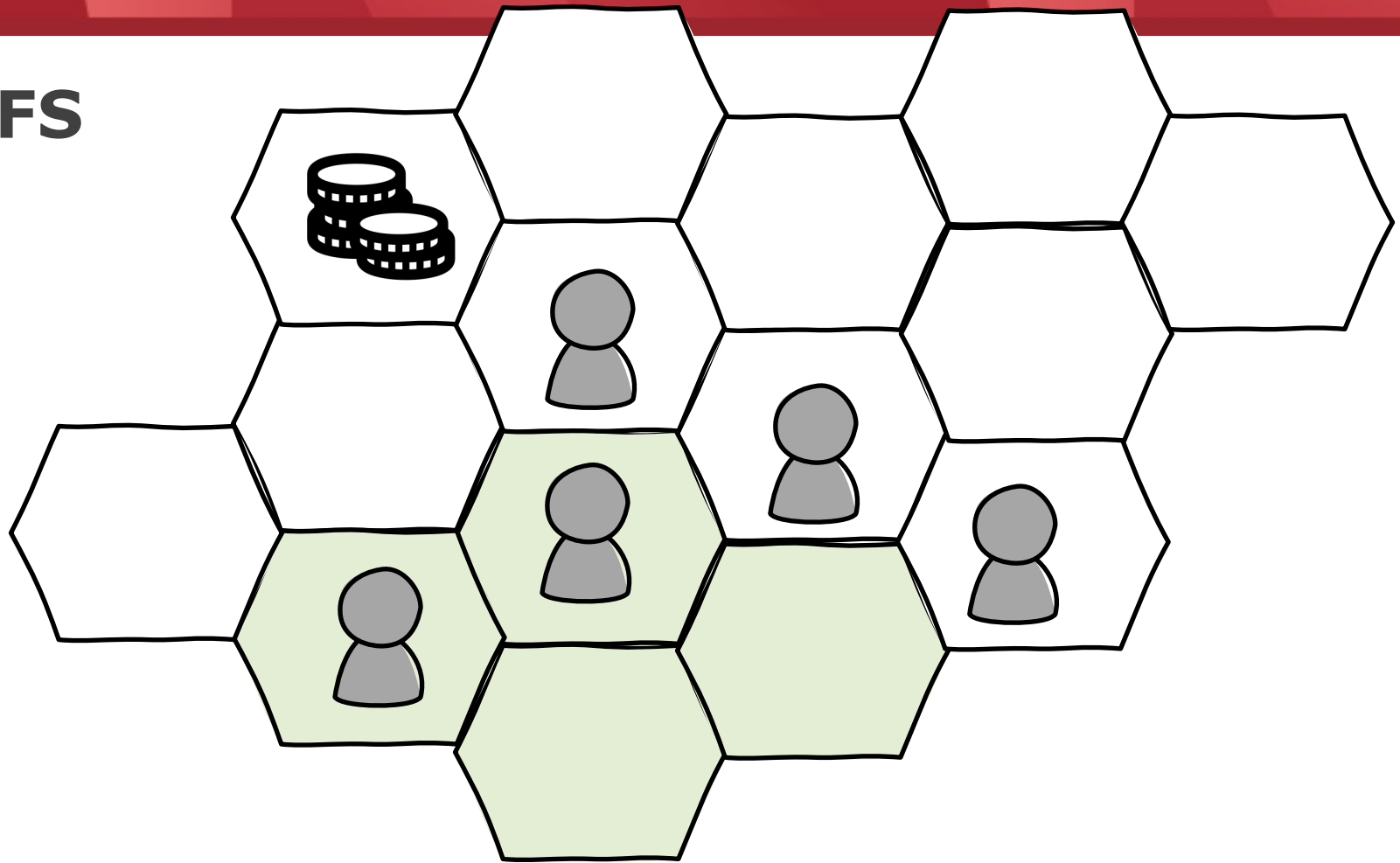


Blind Search: BFS



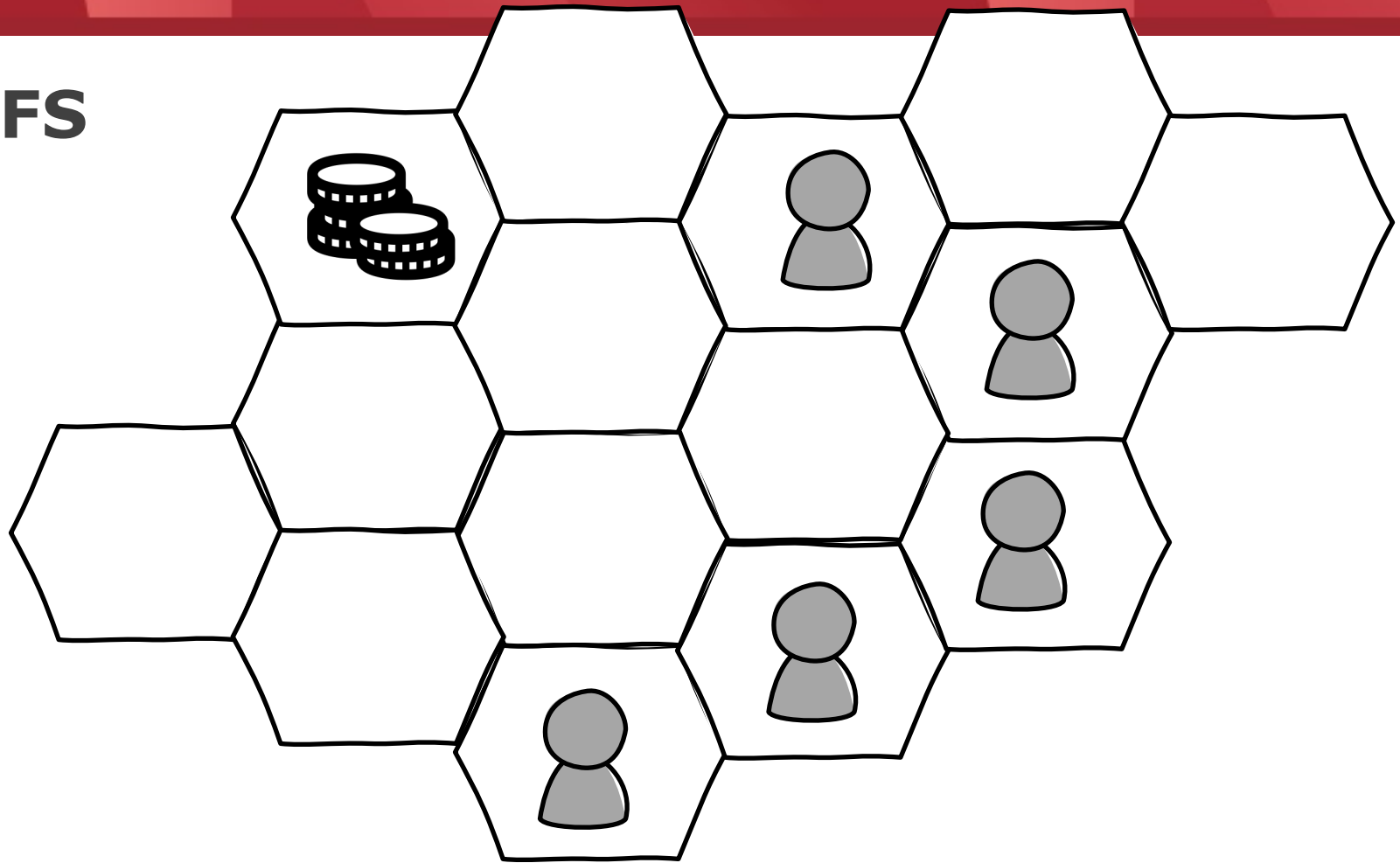


Blind Search: BFS





Blind Search: DFS



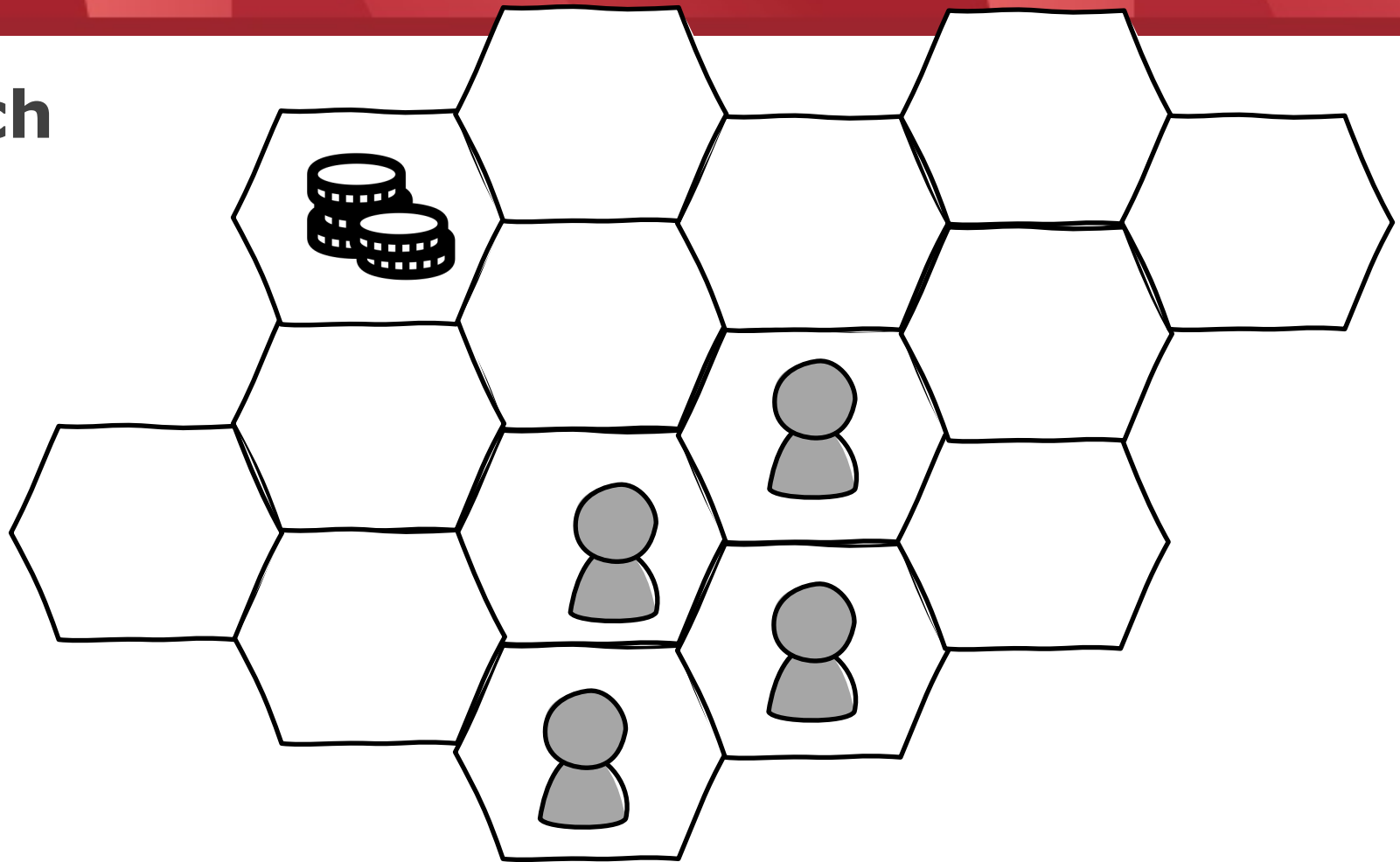
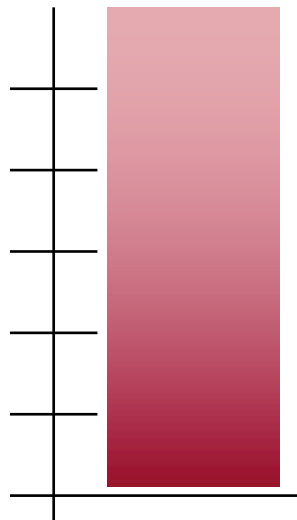


Informed: Measurement Device





Informed Search



Heuristic (hyu-RIS-tik)

- ▶ *Heuriskein*
 - to discover, to find, to search
- ▶ the process of gaining knowledge or some desired result by intelligent guesswork rather than by following some pre-established formula.
 - A function which gives an approximate value (cost estimation) of a solution
- ▶ Heuristic can be contrasted with algorithm



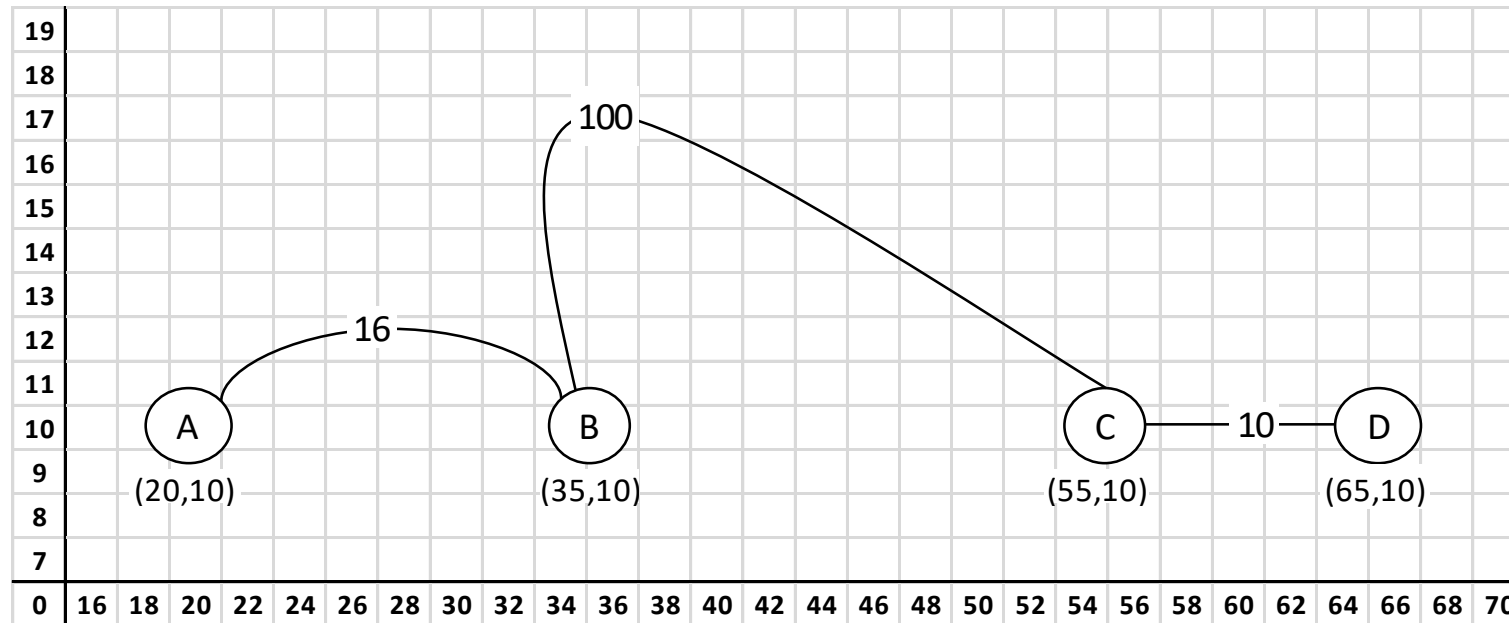
Heuristic Search



Heuristic Function

- ▶ A heuristic function is admissible if the estimated **cost** generated **does not exceed** the actual cost
- ▶ An overestimated heuristic function can make the search process got lost or achieve a non optimal result.
- ▶ A good heuristic function is a function that gives approximate costs that are **close** to the true cost.
- ▶ The closer the actual cost, the better heuristic function it is

Example: Road Distance



With Heuristic function h is the Euclidean distance:

$$d_{ab} = \sqrt{(y_b - y_a)^2 + (x_b - x_a)^2}$$



Example: 8-puzzle game

1	2	3
8	7	4
6		5

Initial State

1	2	3
4	5	6
7	8	

Goal State

Heuristic function $h = ?$



Example: 8-puzzle game

- ▶ 2 possible functions:
 - $h_1(n)$ = number of misplaced tiles
 - $h_2(n)$ = total Manhattan distance (city block distance)
 - # of squares from desired location of each tile



1	2	3
4	5	6
7	8	

Goal State

1	6	8
3	7	
2	4	5

$$h_1(n) = 7$$

$$\begin{aligned} h_2(n) &= 2 + 3 + 3 + 2 + 3 + 2 + 2 \\ &= 17 \end{aligned}$$

1	2	3
8	7	4
6		5

Initial State

$$h_1(n) = 5$$

$$\begin{aligned} h_2(n) &= 2 + 2 + 2 + 3 + 2 \\ &= 11 \end{aligned}$$

Example: Water Jug Problem



Heuristic function $h = ?$

$$h(s) = |y - 2|$$



Heuristic Search

- ▶ Hill Climbing
- ▶ Simulated Annealing (SA)
- ▶ Best-First Search
- ▶ Greedy Best-First Search
- ▶ A*



Heuristic Search

- ▶ Iterative Deepening A* (IDA*)
- ▶ Simplified Memory-Bounded A* (SMA*)
- ▶ Bi-directional A* (BDA*)
- ▶ Modified Bi-directional A* (MBDA*)
- ▶ Dynamic Weighting A* (DWA*)
- ▶ Beam A* (BA*)

Question?





THANK YOU